- Many mathematical statements are constructed by combining one or more propositions. New propositions, called $\boxed{\text{compound propositions}}$, are formed from existing propositions using $\boxed{\text{logical operators}}$. (Refer Boole's 'The Laws of Thought' for further reading)

- **Definition 1:** Let p be a proposition. The $\boxed{\text{negation}}$ of p, denoted by $\neg p$ (also denoted by $\overline{p}$) is the statement "It is not the case that p."

The proposition $\neg p$ is read "not p." The truth value of the negation of p, $\neg p$, is the opposite of the truth value of p.

Example 3: Find the negation of the proposition "Michael's PC runs Linux" and express this in simple English.

Soln: It is not the case that Michael's PC runs Linux/Michael's PC does not run Linux.

Example 4: Find the negation of the proposition "Vandana's smartphone has atleast 32 GiB of memory" and express this in simple English.

Soln: Vandana's smartphone has less than 32 GiB of memory.

Table 1: The Truth Table for the Negation of a Proposition

p	$\neg p$
T	F
F	T

We will now introduce the logical operators that are used to form new propositions from two or more existing propositions. These logical operators are also called $\boxed{\text{connectives}}$.

- **Definition 2:** Let p and q be propositions. The $\boxed{\text{conjunction}}$ of p and q, denoted by $p \wedge q$, is the proposition "p and q." The conjunction $p \wedge q$ is true when both p and q are true and is false otherwise.